

Hani Ahmed

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TECHNICAL SKILLS

Programming & Scripting: C, C++, Python, Shell, PowerShell, Perl, TCL, Java, JavaScript, MATLAB, Regex
Systems & Firmware: Linux/Unix, Windows, BIOS, Firmware, RTOS, Debuggers, Compilers, CPU/APU, PC Architecture
Hardware & RTL: Verilog, System Verilog, VHDL, RTL Design, FPGA, ASIC Design, Physical Design, VLSI, Microcontrollers
Verification & Interfaces: UVM, DFT, PCIe, UCIE, Defect Tracking, Testing, Debugging, Source Control, Perforce, Jira
Software & Data Tools: Git, GitHub, Vim, Docker, FastAPI, SQLite, MySQL, PostgreSQL, XML, HTML, Claude, Gen AI

EDUCATION

Toronto Metropolitan University

Sep 2022 – May 2027

Bachelor of Electrical Engineering

Toronto, ON

- Relevant Coursework: Computer Architecture & Organization, Microprocessor Systems, Digital Systems, Electronic Circuits, Semiconductor Physics, Wireless Communication, Analog Circuits, Probability/Stochastic, Data Structures & Algorithms

EXPERIENCE

FMD Systems Intern

May 2024 – Present

Toronto Metropolitan University Facilities Management & Development

Toronto, ON

- Analyzed 3600+ elevator outage and maintenance data using Python, Pandas, Excel, and Matplotlib to identify reliability trends, recurring faults, and operational improvement opportunities.
- Created and edited technical documentation, Excel reports, and structured stakeholder updates to support troubleshooting, root-cause analysis, project tracking, and data-driven maintenance decisions.

Computer Refurbishment Volunteer

Jun 2023 – Jul 2023

Siemens

Oakville, ON

- Built PCs from individual components including CPUs, memory, storage, graphics cards, motherboards, and power supplies, then installed operating systems, drivers, and basic network configuration.
- Performed system-wide debugging in Windows and Linux/Unix environments by updating BIOS firmware, validating boot readiness, checking BitLocker and secure boot, and troubleshooting hardware/software issues.

PROJECTS

Agentic AI Debug Assistant for RTL and EDA Logs | *Python, FastAPI, RAG, MCP, SQLite, GitHub Actions, Cloud (AWS)*

- Built a cloud-ready agentic Python/FastAPI debug assistant that parses ModelSim-style RTL logs, retrieves RAG context through an MCP server, classifies hardware issues, and generates Markdown debug reports.
- Implemented Regex-based classification and SQLite-backed defect tracking for width mismatches, missing modules, timing violations, syntax errors, and unconnected nets across searchable API issue records.

Mini EDA Tool for Logic Parsing and RTL Generation | *Python, Verilog, EDA Automation, Netlists, Git*

- Built a Python EDA automation tool for ASIC relevant workflows, parsing custom `.logic` files to extract inputs, outputs, gates, internal wires, and gate level netlist relationships.
- Generated synthesizable Verilog RTL for AND, OR, XOR, NOT, NAND, NOR, and MUX structures, translating parsed netlists into reusable hardware description output for chip level design workflows.

Full Custom CMOS D Flip Flop Layout and LVS Verification | *VLSI, Magic, Netgen, Sky130 PDK, ASIC/Physical Design*

- Designed a full custom CMOS D flip flop layout using Sky130 PDK layers, applying ASIC design, physical design, VLSI concepts, and analog/digital transistor behavior through device placement, routing, wells, contacts, vias, clock, and power.
- Created the transistor-level schematic in Xschem and used Magic-extracted layout netlists with Netgen LVS to verify layout/schematic equivalence while debugging device mismatches, connectivity errors, and CMOS pull-up/pull-down behavior.

RISC-V Pipelined Processor RTL Design on FPGA | *SystemVerilog, Verilog, Vivado, FPGA, Chip Level Design*

- Designed a 32-bit RISC-V pipelined processor in Verilog/SystemVerilog within a Linux development environment, implementing PC, register file, ALU, control unit, datapath, and pipeline logic.
- Applied digital systems, computer organization, and logic design fundamentals across fetch, decode, execute, memory, and writeback while debugging branch control, registers, timing, FSMs, flops, and waveform-level HDL behavior.

SerDes Link Training and Lane Repair RTL Simulation | *SystemVerilog, Python, PCIe/UCIe Concepts, RTL Verification*

- Built a SerDes link training simulation using Python lane quality modeling and SystemVerilog RTL control logic to evaluate lane health, detect failed lanes, trigger repair decisions, and remap traffic to spare-lane paths.
- Modeled channel loss, FIR-style equalization, eye opening, and lane quality metrics to connect high-speed SerDes signal behavior with RTL-level control decisions used in PCIe/UCIe-style link reliability flows.
- Verified all single-lane failure cases with directed SystemVerilog testbenches, confirming repair control, failed-lane selection, spare-lane remapping, data integrity, and waveform-level debug behavior.

6502 CPU Emulator in C++ | *C++, Low Level Programming, CPU Architecture, Software Engineering Concepts*

- Built a C++ CPU emulator applying software engineering concepts to implement instruction fetch, decode, registers, memory access, status flags, stack behavior, and addressing modes.
- Debugged instruction execution and control-flow behavior using structured tests, trace outputs, and hardware architecture concepts related to CPUs, embedded systems, and system software.

Areas for Growth: Machine Learning for EDA, semiconductor device physics, chip-level ASIC integration, and physical design process